

JEE MAIN REVISION MASTER: IONIC EQUILIBRIUM

Subject: Chemistry | Target Scope: JEE Main Previous Years | Empirical Base: 8 Analyzed PYQs

1. High-Yield Concept Priority Matrix

Priority Tier	Concept Name	Frequency	Category
Moderate	pH Scale, Ostwald's Law & Buffer Solutions	5 PYQs	Ionic Equilibria
Moderate	Solubility Product (Ksp) & Salt Hydrolysis	3 PYQs	Precipitation & Hydrolysis

2. Master Formula Reference Card

Concept / Topic	Mathematical Formulation & Governing Equations
pH Scale, Ostwald's Law & Buffer Solutions	$\text{pH} = \text{pK}_a + \log_{10} \frac{[\text{Conjugate Base}]}{[\text{Acid}]}, \quad [\text{H}^+] = \sqrt{K_a \cdot c}, \quad K_w = 10^{-14}$
Solubility Product (Ksp) & Salt Hydrolysis	$\text{For } A_x B_y: K_{sp} = x^x y^y S^{x+y}, \quad Q > K_{sp} \implies \text{Precipitation}, \quad \text{Weak-Weak: } \text{pH} = 7 + \frac{1}{2}(\text{pK}_a - \text{pK}_b)$

3. Deep-Dive Concepts & Linked PYQ Evidence

3.1 pH Scale, Ostwald's Law & Buffer Solutions (Moderate Yield — 5 PYQs)

Core Principle: pH calculations: $\text{pH} = -\log[\text{H}^+]$, $\text{pH} + \text{pOH} = 14$ at 25°C . Ostwald's dilution law for weak acid: $\alpha = \sqrt{(K_a / c)}$, $[\text{H}^+] = \sqrt{K_a \cdot c}$. Henderson-Hasselbalch equation for acidic buffer: $\text{pH} = \text{pK}_a + \log([\text{Salt}] / [\text{Acid}])$. Basic buffer: $\text{pOH} = \text{pK}_b + \log([\text{Salt}] / [\text{Base}])$.

■ **Caution / JEE Pitfall:** A 10^{-8} M HCl solution does NOT have $\text{pH} = 8$ (it is acidic!). Accounting for water dissociation $[\text{H}^+] = 10^{-8} + 10^{-7} = 1.1 \times 10^{-7}$ implies $\text{pH} \approx 6.96$.

■ **Speed Trick:** When $[\text{Salt}] = [\text{Acid}]$ in a buffer, maximum buffer capacity is achieved and $\text{pH} = \text{pK}_a$.

Representative Source PYQs:

- [2026 Shift] Q#1 (MCQ): "A 0.15 mole of pyridinium chloride has been added to 500cm³ of 0.2 M pyridine solution (a base). Assuming there is no change..." [Ans: N/A]
- [2022 Shift] Q#4 (MCQ): "Which among the following has highest pH ?..." [Ans: N/A]
- [2021 Shift] Q#1 (MCQ): " K_a values for acids H_2SO_3 , HNO_2 , CH_3COOH and HCN are respectively 13×10^{-2} , 4×10^{-4} , 1.8×10^{-5} ..." [Ans: N/A]

3.2 Solubility Product (K_{sp}) & Salt Hydrolysis (Moderate Yield — 3 PYQs)

Core Principle: Solubility product K_{sp} : For AB salt $K_{sp} = S^2$; for AB₂ salt $K_{sp} = 4S^3$; for A₂B₃ salt $K_{sp} = 108S^5$. Precipitation occurs when ionic product $Q_{ip} > K_{sp}$. Salt hydrolysis: Salt of weak acid + strong base (ext CH₃COONa) gives alkaline solution with $pH = 7 + (1/2)(pK_a + \log C) > 7$.

■ **Caution / JEE Pitfall:** Solubility of a sparingly soluble salt (like AgCl) decreases significantly in the presence of a common ion (like NaCl) due to Le Chatelier's shift.

■ **Speed Trick:** For AB₂ salt (e.g. CaCl₂, Mg(OH)₂): $S = 3\sqrt{\frac{K_{sp}}{4}}$.

Representative Source PYQs:

- [2024 Shift] Q#2 (MCQ): "Solubility product of CaC_2O_4 at a given temperature in pure water is $4 \times 10^{-9}(\text{molL}^{-1})^2$. Solubility of CaC_2O_4 ..." [Ans: N/A]
- [2023 Shift] Q#3 (MCQ): "A weak acid with pK_a 5.9 and weak base with pK_b 5.8 are mixed in equal proportions. pH of the resulting solution is..." [Ans: N/A]
- [2019 Shift] Q#4 (MCQ): "Solubility of AgCl is least in..." [Ans: N/A]

4. Benchmark Illustrative PYQs with Master Solutions

Benchmark Problem 1: KCET 2026 (Q#1)

A 0.15 mole of pyridinium chloride has been added to 500 cm³ of 0.2 M pyridine solution (a base). Assuming there is no change in volume upon mixing (Given: pK_b of pyridine = 8.82), the pH of the resulting solution is

- (A) 5
- (B) 6
- (C) 7
- (D) 8

Correct Answer: None

Solution Analysis: Moles of pyridine present initially: $0.2 \times (500 / 1000) = 0.1 \text{ mol}$ Moles of pyridinium chloride added: 0.15 mol Pyridine is a weak base, and pyridinium chloride gives its conjugate acid, pyridinium ion. So the mixture acts as a basic buffer. For a basic buffer, $pOH = pK_b + \log([salt] / [base])$ Since volume is the same for both, we can directly use moles: $pOH = 8.82 + \log(0.15 / 0.10)$ $pOH = 8.82 + \log(1.5)$ $\log(1.5) \approx 0.18$ So, $pOH = 8.82 + 0.18 = 9.00$ Now, $pH = 14 - pOH$ $pH = 14 - 9 = 5$ So the pH of the result

Benchmark Problem 2: KCET 2024 (Q#2)

Solubility product of CaC_2O_4 at a given temperature in pure water is $4 \times 10^{-9}(\text{molL}^{-1})^2$. Solubility of CaC_2O_4 at the same temperature is

- (A) $6.3 \times 10^{-5} \text{ molL}^{-1}$
- (B) $2 \times 10^{-5} \text{ molL}^{-1}$
- (C) $2 \times 10^{-4} \text{ molL}^{-1}$
- (D) $6.3 \times 10^{-4} \text{ molL}^{-1}$

Correct Answer: None

Solution Analysis: Given, $K_{sp} = 4 \times 10^{-9}(\text{molL}^{-1})^2$ $S = ?$

$\text{CaC}_2\text{O}_4 \rightleftharpoons \text{Ca}^{2+} + \text{C}_2\text{O}_4^{2-}$ $4 \times 10^{-9} = S^2 \Rightarrow S = \sqrt{4 \times 10^{-9}} = 6.3 \times 10^{-5} \text{ mol/L}$
Thus, the solubility of CaC_2O_4 is $6.3 \times 10^{-5} \text{ mol/L}$.